

ClaimSense 360: Ultimate 34-Module Defense & Technical Viva Encyclopedia

COMPLETE ANSWERS TO ALL 34 EXAMINER QUESTION CATEGORIES & ARCHITECTURE PROOFS

Executive Elevator Pitch: 'ClaimSense 360 is an Enterprise Multi-Modal AI Motor Insurance Claims Platform. It solves India's ₹45,000 Crore annual fraud problem by combining Tabular XGBoost ML, SHAP Game-Theoretic Explainability, TF-IDF Narrative Analysis, YOLOv8 + ResNet-18 Computer Vision, and Digital Image EXIF Forensics in under 0.5 seconds.'

1. Multi-Modal AI Ensemble & Math Specifications

XGBoost Fraud Classifier: Trained on 1,000 Kaggle claims across 46 financial/driver features.

SHAP Game Theory: $\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \left[\frac{|S|!(|N|-|S|-1)!}{|N|!} \right] * [v(S \cup \{i\}) - v(S)]$

Computer Vision Damage Score: $Severity = 0.50 * S_{YOLO} + 30 * C_{ResNet} + 20 * E_{Edge}$

PIL EXIF Image Forensics: Detects missing camera headers on web/downloaded photos, tracking forensic_penalty = 18.5 without corrupting pure XGBoost fraud score.

2. Full-Stack Technology Justification Matrix

Layer / Module	Technology Used	Selection Justification
Frontend UI	Next.js 15.1 + React 19	Server-Side Rendering, App Router, type-safe Next proxy
Design Tokens	Tailwind CSS v3.4 + Framer Motion	HSL design variables (#173B32, #E66A4E), fluid state motion
Backend REST API	FastAPI 0.110 + Uvicorn	Sub-millisecond ASGI performance, auto Swagger OpenAPI
Database & ORM	SQLite 3 + SQLAlchemy 2.0	Zero-config local relational DB, lightweight ORM schema
Tabular Fraud ML	XGBoost 2.0 Classifier	Handles non-linear financial ratios and driver interactions
Explainable AI	SHAP TreeExplainer	Game-theoretic local feature attribution (IRDAI compliant)
Computer Vision	YOLOv8 + PyTorch ResNet-18	22-class vehicle damage detection + deep feature vector norm
Image Forensics	PIL Telemetry Inspector	Raw EXIF camera header scanning; flags Google downloads

3. Examiner Verification: 'Is your AI actual or fake/dummy?'

Your Answer: 'Sir, our AI engine is 100% real and verifiable. We have live trained fraud_model.joblib (XGBoost), shap.TreeExplainer, yolov8n.pt PyTorch weights, and PIL EXIF telemetry. When you submit a new claim or upload a downloaded image, the model computes real-time SHAP factor contributions and renders a live Image Forensics Warning badge. We can run our Python test verification script scratch/verify_yolo_execution.py right now to prove live GPU tensor execution!'